Orthogonal projection

$A \in M_{n \times m}(F)$, $b \in F^n$

We want to solve

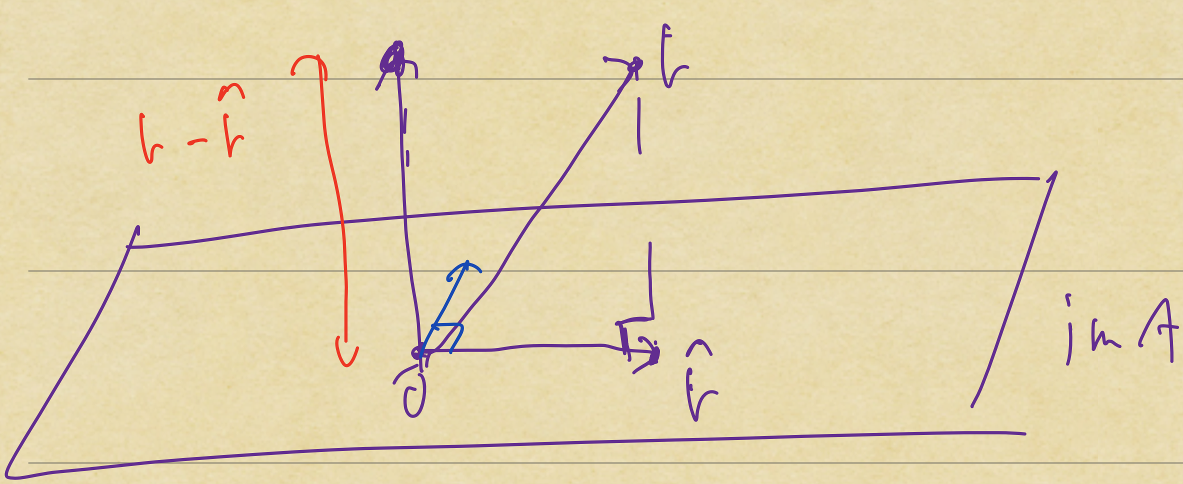
$$Ax = b$$

$x \in F^m$

Only possible if $b \in \operatorname{im} A \leq F^n$

Q: What to do when $b \notin \operatorname{im} A$?

In this how can we find x s.t. Ax best approximates b?



A: We want to first choose $\hat{r} \in \operatorname{im} A$ s.t. $r - \hat{r}$ is orthogonal to the image of A

Def^n $W \leq F^n$: subspace

$$W^{\perp} = \{ v \in F^n : v \cdot w = 0, \forall w \in W \}$$

"W-perp" $W^{\perp}$ is the orthogonal complement of W in F^n

In this language, we are asking for

$$r - \hat{r} \in (\operatorname{im} A)^{\perp}$$

Observation:

$$(\operatorname{im} A)^{\perp} = \ker A^T$$

Pf:

$$(\operatorname{im} A)^{\perp} = \{ v \in F^n : v \cdot w = 0, \forall w \in \operatorname{im} A \}$$

$$= \{v \in F^n : v \cdot Au = 0, \ \forall u \in F^m\}$$

$$= \{v \in F^n : v^T A u = 0, \ \forall u \in F^m\}$$

Fact ②

$$= \{v \in F^n : (A^T v)^T u = 0, \ \forall u \in F^m\}$$

$$= \{v \in F^n : \underbrace{(A^T v)}_{\in F^m} \cdot u = 0, \ \forall u \in F^m\}$$

$$= \{v \in F^n : A^T v = 0\}$$

$$= \ker A^T$$

$$\boxed{\begin{gathered}(AB)^T = B^T A^T \\ (A^T)^T = A\end{gathered}}$$

Facts:

① $(F^n)^\perp = \{0\}$

Pf: $\overbrace{\begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}}^{v} \cdot e_i = a_i$

$\Rightarrow$ If $v \cdot e_i = 0, \ \forall i$

$\Rightarrow$ $v = 0$.

② $A \in M_{n \times m}(F)$, $u \in F^m$, $v \in F^n$

$$v \cdot Au = (A^T v) \cdot u$$

Second reformulation

$$b - \hat{b} \in \ker A^T$$

$$\iff A^T(b - \hat{b}) = 0$$

$$\iff \underline{A^T b = A^T \hat{b}}.$$

$$+ \quad \hat{b} \in \operatorname{im} A \iff \underline{\hat{b} = Au}, \quad u \in F^m$$

Def^n A **least squares approximate** solution to $Ax = b$ is a solution to

$$\boxed{\underbrace{A^T A}_{m \times m} x = A^T b}$$

$\boxed{F = \mathbb{R}}$

Def$\underline{^n}$ $W \leq \mathbb{R}^n$

Then the orthogonal projection

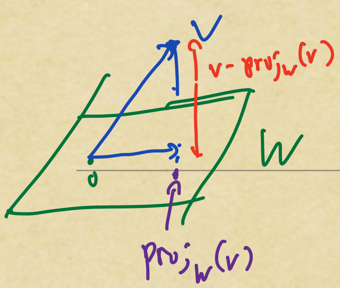
$$\text{proj}_W : \mathbb{R}^n \longrightarrow \mathbb{R}^n$$

is the unique linear transformation with the following properties:

1. $\text{proj}_W \circ \text{proj}_W = \text{proj}_W$

2. $\text{im}\ \text{proj}_W = W$

3. $\forall v \in \mathbb{R}^n$



$$\text{proj}_W(v) \cdot (v - \text{proj}_W(v)) = 0$$

Fact ①: In this situation,

$$\begin{aligned}
&\mathrm{proj}_W(v - \mathrm{proj}_W(v)) \\
&= \mathrm{proj}_W(v) - \mathrm{proj}_W(\mathrm{proj}_W(v)) \\
&= \mathrm{proj}_W(v) - \mathrm{proj}_W(v) \\
&= 0
\end{aligned}$$

Def^n $W \leq \mathbb{R}^n$

An orthonormal basis for W is a basis $(u_1, \ldots, u_m)$ for W with the following properties:

① $u_i \cdot u_j = 0$ if $i \neq j$

② $u_i \cdot u_i = 1$ $\forall i$

Example: $W = \mathbb{R}^n$ then $(e_1, \ldots, e_n)$ is an orthonormal basis for $\mathbb{R}^n$.

Proposition: Suppose that

$(u_1, \ldots, u_m)$ is an orthonormal basis

for $W \leq \mathbb{R}^n$

Then

$$\text{proj}_W(v) = (v \cdot u_1)u_1 + \ldots + (v \cdot u_m)u_m$$

gives the orthogonal projection onto

$W \leq \mathbb{R}^n$